



USN

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Seventh Semester B.Tech (CBCS) Degree Examinations

December 2025

ME2342-1 – INTERNAL COMBUSTION ENGINES

(For ME)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **one full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. The term stoichiometric mixture refers to
 - A) Lean mixture
 - B) Chemically correct air-fuel ratio
 - C) Rich mixture
 - D) Maximum power-producing mixture
2. The afterburning phase occurs when
 - A) All fuel burns in the ignition delay
 - B) There is no air-fuel mixture
 - C) Combustion continues into the expansion stroke
 - D) Injection stops immediately
3. In a modern SI engine, which ignition system is most commonly used?
 - A) Magneto ignition system
 - B) Electronic ignition system
 - C) Battery ignition system
 - D) Hot tube ignition system
4. Combustion in SI engine takes place in how many stages?
 - A) 2
 - B) 4
 - C) 3
 - D) 5
5. Pre-combustion chambers are typically used in
 - A) Gasoline engines
 - B) Indirect injection diesel engines
 - C) Direct injection engines
 - D) Carbureted engines
6. In a diesel engine possibility of knocking is at the
 - A) End of combustion
 - B) Beginning of combustion
 - C) During after burning
 - D) Any stage of combustion
7. The time between the start of injection and actual start of combustion is called
 - A) After-burning
 - B) Delay period
 - C) Flame propagation period
 - D) Ignition lag
8. Which type of mixture is present in CI engine before ignition?
 - A) Heterogeneous mixture
 - B) Homogeneous mixture
 - C) Pure fuel vapour
 - D) Stoichiometric mixture
9. CO is formed primarily due to
 - A) Rich air-fuel mixture
 - B) Lean air-fuel mixture
 - C) High temperature combustion
 - D) Stoichiometric combustion
10. Smoke in diesel engines is caused by
 - A) Incomplete combustion of fuel
 - B) Overheating
 - C) Lean mixture
 - D) Water injection
11. LPG is stored under which condition to keep it in liquid form?
 - A) Low temperature
 - B) High temperature
 - C) Low pressure
 - D) High pressure
12. Alcohol fuels are generally derived from
 - A) Electrolysis
 - B) Fossil fuels
 - C) Oil refining
 - D) Biomass fermentation

13. Which alternative fuel has the highest-octane rating?
 A) Ethanol B) CNG
 C) LPG D) Hydrogen
14. Biodiesel is produced by transesterification of
 A) Vegetable oils or animal fats B) Coal
 C) Natural gas D) Petroleum
15. Photochemical smog is mainly caused by
 A) NO_x and unburnt hydrocarbons (HC) B) SO₂ and particulate matter
 C) Lead particles D) CO and CO₂
16. Exhaust Gas Recirculation (EGR) is mainly used to control
 A) CO emissions B) HC emissions
 C) Smoke emissions D) NO_x emissions
17. Fuel cells produce electricity by
 A) Mechanical energy B) Combustion
 C) Solar conversion D) Electrochemical reaction
18. The Bosch smoke meter is used to measure
 A) Engine temperature B) Fuel consumption
 C) Cylinder pressure D) Smoke opacity in diesel exhaust
19. The most commonly used battery in modern electric vehicles is
 A) Lithium-ion battery B) Lead-acid battery
 C) Nickel-Metal Hydride (NiMH) battery D) Nickel-Cadmium (Ni-Cd) battery
20. A lean burn engine operates with
 A) Excess air (lean mixture) B) Rich fuel-air mixture
 C) Pure fuel injection D) Stoichiometric fuel-air mixture

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit - I		Marks	BT*	CO*	PO*
1. a)	Explain the stages of combustion in SI engines with the help of a neat pressure-crank angle diagram. How is it different from the ideal diagram?	8	L*2	1	1
b)	What is ignition delay in CI engines? Explain the physical and chemical processes involved in ignition delay and the factors affecting it.	8	L1	2	1
2. a)	What is flame speed? Discuss the important factors affecting flame speed in SI engines combustion.	8	L2	1	1
b)	What is swirl in diesel engine combustion? Explain how swirl, squish, and turbulence are generated in the combustion chamber and their effect on combustion.	8	L2	2	1
3. a)	What is pre-ignition? State its causes, effects on engine performance, and methods of prevention.	8	L1	1	1
b)	Describe the different types of fuel injector nozzles used for diesel fuel combustion in the CI engine with neat sketches.	8	L2	2	1
Unit - II					
4. a)	What are the major pollutants emitted from SI and CI engines? Explain their sources and effects on environment and human health.	8	L1	3	1
b)	Discuss the advantages and limitations of CNG as an automotive fuel.	8	L2	4	1
5. a)	Discuss the effects of design and operating variables on exhaust emissions of a SI engine.	8	L2	3	1
b)	Explain the different methods of onboard storage of hydrogen for IC engine use.	8	L2	4	1

ME2342-1

SEE - December 2025

- | | | | | | |
|-------|--|---|----|---|---|
| 6. a) | What are the main sources of HC emission in petrol engine? Explain briefly. | 8 | L1 | 3 | 1 |
| b) | Explain the process of methanol production from municipal waste and ethanol from sugar cane. | 8 | L2 | 4 | 1 |

Unit – III

- | | | | | | |
|-------|---|---|----|---|---|
| 7. a) | What are lean burn engines? State the advantages and limitations of lean burn operation in SI engines. | 8 | L1 | 5 | 1 |
| b) | What is a stratified charge engine? How does it differ from a homogeneous charge engine? | 8 | L1 | 5 | 1 |
| 8. a) | What is meant by electric drive? Explain with neat block diagram components of an electric drive system. | 8 | L2 | 5 | 1 |
| b) | What is a fuel cell? How it differs from battery? Explain working of a hydrogen fuel cell with a neat sketch. | 8 | L1 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
